# Speech-Eval

## Contrastive Speech Delivery Analytics with Temporal Flaw Grounding

Multimodal AI Hackathon 2026 - Track C | October 2026

## 1. Problem and approach

Speech-Eval shows you delivery problems such as speaking too fast, too quietly, or pausing in an awkward place. It marks the affected words and times so you can listen to the exact part. It explains each flag using measured differences from reference readers or your earlier take.

Frozen A: precision 0.125, recall 0.400, F1 0.190.
Frozen B: precision 0.018, recall 0.390, F1 0.035.
A clean alarms: 1.341/min; dev guard failed (1.777 versus <=0.5/min).
Release: 1,103 indexed recordings, 36.3 h; repeated scientific hashes match.
Limits: synthetic edits, low precision, no genuine-take or independent-rater validation.

Content quality is out of scope. Controlled edits supply reproducible operation labels, not independent judgments of good speaking. A uses same-text train readers; B uses the speaker's earlier take; C-lite uses sentence norms without a same-text reference. Key-moment messages are labelled heuristics, not scored judgments.

## 2. Contrastive dataset construction

Sources: 15 Gettysburg and 13 O Captain LibriVox readings, plus one JFK Rice and one Reagan Challenger recording. USA public-domain evidence is recorded in the release manifests and dataset card; automatic evidence checks are not human legal review. Raw hashes remain immutable.

Canonical/as-spoken differences were checked with local CTC candidate likelihoods calibrated on 300 controls, empirical fake acceptance ≤2.0%. This in-sample proxy is not listening or guaranteed accuracy on real disagreements. Unsettled words and dependent evidence remain masked. Number expansion retains index mappings. Acoustic audit.

MFA/WhisperX were compared on 37 words / 74 edges after 23 unclear-word exclusions. **Claude placed visual boundaries without listening**; independence from model templates was asserted, not independently established. Human spot-check is pending. Pooled median differences: MFA 52.0 ms, WhisperX 72.0 ms; MFA is primary. Both passed independent CPU rerun identity. These are differences from a limited visual reference, not established human accuracy. Bake-off.

Eligible sources receive seven effects at five edit strengths plus mixed variants, with seeded placement. One clipped O Captain source is audit-only. The release has 1,103 indexed variants / 36.3 h: 30 originals and 1,073 edited recordings. Raw downloads/duplicate rates are not counted again. Train/dev/test have 18 / 4 / 7 speakers. Measured counts.

| Type | Controlled operation | Increasing edit strength |
|---|---|---|
| rushing / dragging | Pitch-preserving stretch | Raise / lower speed factor |
| monotone | PSOLA around median pitch | Retain less pitch range |
| too_quiet | Flat/trailing gain | More attenuation |
| awkward_pause | Phrase-internal room tone | Longer inserted gap |

| Type | Controlled operation | Increasing edit strength |
|---|---|---|
| missing_pause | Shorten existing gap | Lower retained fraction |
| mumbling | Zero-phase low-pass plus gain | Lower cutoff/gain |

Parameters are in configs/flaws.yaml. Time maps transport source indices into the flawed timeline; labels bind type, edit-strength severity and interval. Real MFA is run separately for detection. Oracle alignments serve label/QA only; edit boundaries are not guaranteed perceptual boundaries. Genuine takes are pending.

## 3. Features and speaker normalisation

Praat pitch/intensity/HNR and spectral/MFCC tracks feed word, short-window and sentence aggregates. Speaker-relative transformations reduce voice/microphone offsets, not all differences:

$st(t) = 12 \log_2[f_0(t) / \text{median } f_0]$; $Lrel(t) = L(t) - \text{median } L$.

Medians use voiced frames inside aligned words. Rate (syllables/second) is not speaker-normalised. Clarity is a high-band energy ratio, not intelligibility. Comparison rate/pitch-variation/pause features are log-transformed; loudness and clarity remain dB. Uncertain-word masks exclude dependent aggregates. Feature normalisation does not silently edit stored loudness.

## 4. Detection, grounding and explanations

Clean train readers give word/feature median mu and robust spread $s = 1.4826 \text{ median}(|x - mu|)$. Pooled train spread and shrinkage $n0 = 5$ give:

$\sigma^2_{i,f} = (n\, s^2_{i,f} + n_0\, \sigma^2_{pool,f}) / (n + n_0)$; $z_{j,f} = (x_{j,f} - \mu_{i,f}) / \sigma_{i,f}$.

Whole-word mapping excludes unmatched/partial correspondences. Runs within sentences become regions; pauses are boundary events. Directional feature rules and contradictory-pair suppression are explicit. Frozen selected k is **3.50 for every type**, not the draft default 2.0. Dev severity cut points are [2.0, 2.75, 3.5, 4.059, 6591.0]; matched dev Spearman is 0.16. The unusual final cut point/weak correlation are retained, not repaired with test or user takes. Calibration.

Templates quote measured participant/reference deviations and flagged words. The following is **an example of real system output**, quoted verbatim from analysis 86964ad39ead, Mode A test file gettysburg__lv_sa__awkward_pause_s3_01:

> 76.0–77.6 s, "it can": a 1.6 s pause between "it" and "can" where good readers pause 6.9e-18 s (1.6 s longer, 6e+03 SD above). It breaks the sentence.

The detected interval is 75.990-77.580 s, participant pause 1.590 s, reference pause 6.939e-18 s, z = 6046.0. Its matching injected label spans 76.035-77.535 s in the label file. It is a correct hit under the frozen pause-centre rule. The extreme z reflects near-zero reference spread; it is not evidence of that much perceptual severity. This existing test example is for report illustration, not v2 tuning. Measured rule explanations are **not causal proof**.

B uses the own earlier take; JFK/Reagan test originals are within-speaker references, not population training/calibration. C checks sentence rate, pitch range, final loudness drop and non-final end pause; mumbling/awkward pause are unsupported. Number/end guidelines never alter scores.

## 5. Scoring and validation

For first-to-last-word duration D, duration penalty is sum(v × d/D × 10); pause penalty is sum(0.4 × v). Clip 10 - penalty to [0,10], decimal half-up round, then weight rounded categories: pace/pitch 0.25 each, pauses 0.20, volume/clarity 0.15 each. These are configurable design choices, not an external standard. Severity-3 rushing over 10.0% gives pace 7.0. Validation.

The report gives per-type, not aggregate, monotonicity:

| Type | Median severity-score Spearman | Non-increasing % | Constant series |
|---|---|---|---|
| awkward_pause | -0.87 | 85.7 | 3 |
| dragging | -0.71 | 71.4 | 2 |
| missing_pause | -0.71 | 100.0 | 4 |
| monotone | -0.71 | 28.6 | 0 |
| mumbling | -0.89 | 71.4 | 0 |
| rushing | -0.71 | 71.4 | 1 |
| too_quiet | -0.71 | 85.7 | 3 |

Each type has seven complete series. Constant ranks have undefined Spearman; non-increasing alone is not sensitivity. **Human agreement is pending.** The blind pack/scripts exist but no real rater responses. The form asks overall delivery 1–5, flaw type(s) and approximate time, not five category ratings. Scripts compute system–mean-human/per-rater Spearman, mean defined pairwise human Spearman and exact flaw-set agreement with denominators; no fabricated responses or CIs.

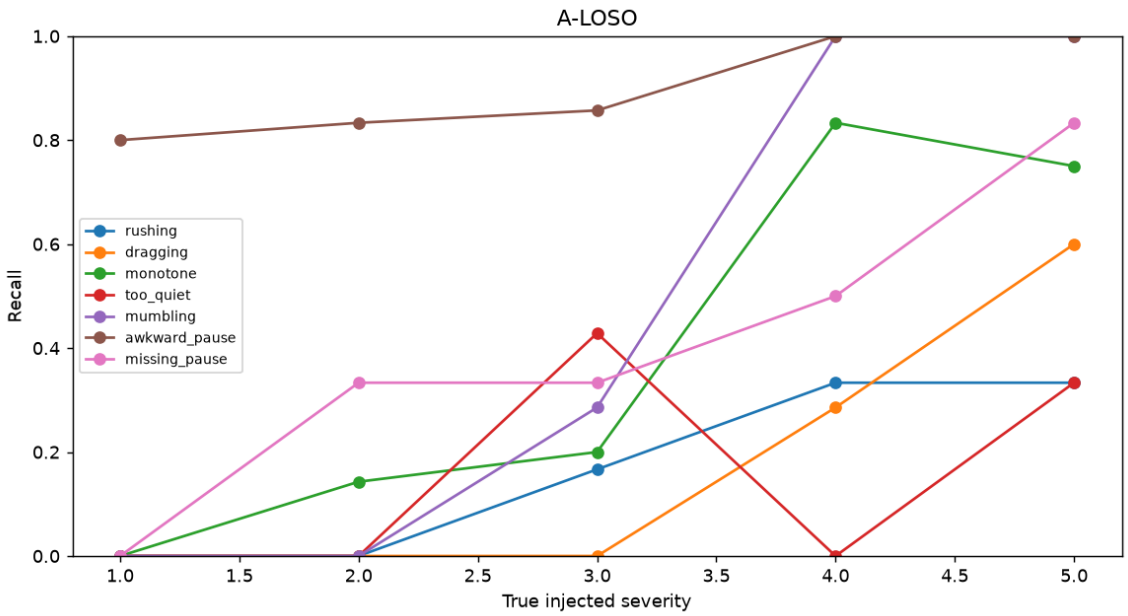
Frozen host two-run identity: 10/10. Later fresh-clone CPU Docker reproductions matched all 563 scientific outputs twice, including cross-environment checks. Scope and SHA manifests.

## 6. Results across settings

Thresholds used dev only. Original final test commit: aaf1832fe24a193d4c90f9e7a2ff9537b753335c. Later test runs are **reproductions**, not tuning. Same-type one-to-one region IoU ≥0.3; pause centre distance ≤0.25 s. Frozen results.

| Setting | Files | Precision | Recall | F1 [95% CI] | Clean alarms/min |
|---|---|---|---|---|---|
| A-LOSO | 190 | 0.125 | 0.400 | 0.190 [0.156, 0.223] | 1.341 |
| B-same-speaker | 76 | 0.018 | 0.390 | 0.035 [0.026, 0.045] | 0.000 |

A highest recall: awkward_pause 0.903; lowest: rushing 0.167. Matched overall onset/offset differences are 26.0 / 40.5 ms (A), 31.0 / 35.5 ms (B), conditional on matches, not all truth regions. Dev clean-alarm guard **failed**: 1.777 versus ≤0.5. Learned detector was not trained. No pooled severity-1/2 or severity-4/5 recall is inferred from the chart without a reported numeric summary.




A-LOSO recall versus injected severity

## 7. No-reference comparison

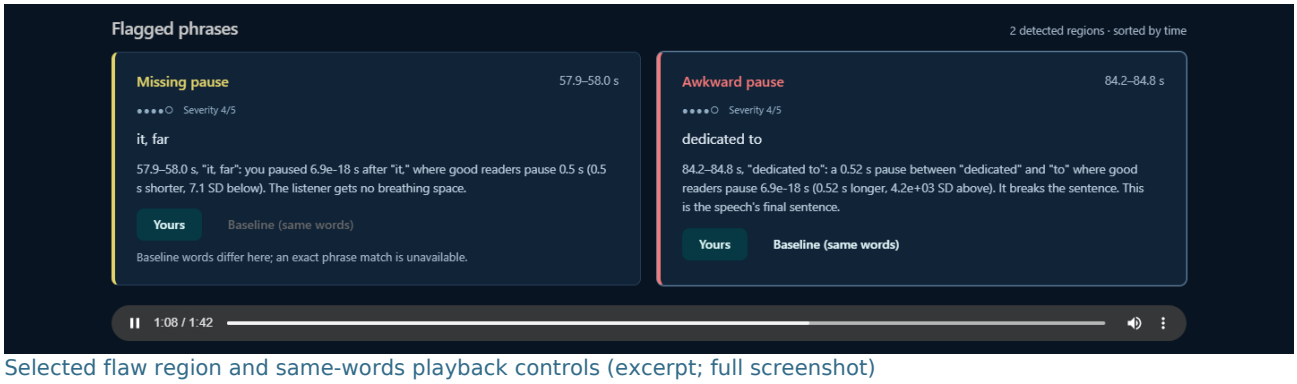
C uses **different eligibility and sentence/type matching**. Its paired comparison on 175 files stays separate from Phase6:

| Paired setting | Precision | Recall | Clean alarms/min |
|---|---|---|---|
| Same-text / own-original reference | 0.106 | 0.324 | 0.584 |
| C-lite norms | 0.020 | 0.179 | 4.159 |

C too-quiet truth is trailing-only; unsupported single effects are omitted and mixed files score supported types only. Lower aggregate recall/more clean alarms describe controlled edits, not humans. Mode C. Genuine take accuracy remains pending. The real-MFA **synthetic workflow fixture** missed its injected region in A and B; no tuning followed. Phase10.

## 8. Dashboard

The results view combines the waveform, flagged spans, rubric scores and generated explanations. Selecting a region isolates its words and supports participant and same-words baseline playback. Screenshots are from the verified real dashboard, not new performance measurements. Browser checks.




Selected flaw region and same-words playback controls (excerpt; full screenshot)



Dashboard results: waveform and score card (excerpt; full screenshot)

## 9. Limitations and future work

Controlled edits are not real co-occurring flaws, and neutral reading is not ideal oratory. Few texts/speakers and English-only proxies limit generalisation. The unheard visual reference and unclear-word exclusions introduce uncertainty and selection bias; model-to-visual differences do not establish annotation accuracy. C/key-moment heuristics and rubric scores lack human validation. Low precision, the failed dev guard, constant/non-monotonic series and fixture miss remain limits. Future work needs consented recordings and independent labels/ratings; detector changes must be labelled post-hoc, not presented as frozen evaluation.

## 10. Reproducibility and attributions

make reproduce-quick DATA_ARCHIVE=/absolute/path/speech-eval-dataset-v1.zip imports the separately supplied archive and runs one text/three test readers. make reproduce re-verifies inputs and reproduces frozen Phase6/7/8 numbers in pinned CPU Docker, not every historical ASR/excerpt decision or human/video work. Quick: 11.829 / 11.954 seconds; full: 225.204 / 223.142 seconds after import/build. Hashes: data/release/reproduction_hashes.json and reproduction report. The narrow legacy-fixture gate exemption is a disclosed post-hoc release adapter, not a detector change; other historical inputs/code are verified.

Audio attribution: LibriVox volunteers and JFK/Reagan institutional collections; source-specific USA public-domain evidence. Full-precision scientific values remain in the original JSON; display rounding does not change them.

Pending items

Genuine takes, consent/as-spoken notes, independent ratings and human alignment spot-check.

Archive upload URL, author confirmation and recorded/published video URL.

Owner-confirmed formal code/annotation licence files; total disk requirement.

Unreported aggregate human/severity/C-F1 and pitch-improvement values remain unclaimed.

See NEXT and placeholder inventory.